1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

In muscle-invasive bladder cancer (MIBC), high intratumoral TIL density indicates immune presence but does not guarantee response to PD-1/PD-L1 blockade because S100A9 signaling from tumor and myeloid cells drives adaptive upregulation of PD-L1 and induces T cell exhaustion. Consequently, TIL-high tumors that are also S100A9-high and PD-L1-high will exhibit reduced objective therapy_response to checkpoint inhibitors despite high bTMB or MSI/dMMR, whereas TIL-high tumors with low S100A9/PD-L1 will have a higher likelihood of benefit; urine-DNA methylation biomarkers may serve as a noninvasive surrogate to monitor and predict this mechanistic state and response trajectory.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	S100A9, PD-L1
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

S100A9: S100A9 → PD-L1 → bTMB axis (Core Path 1) provides a mechanistic bridge linking tumor immunogenicity to adaptive immune suppression via an inflammatory mediator; (b) MSI/dMMR (Core Path 3) and urine-DNA methylation (Core Path 2) inform which tumors are intrinsically more immunogenic or amenable to noninvasive monitoring, but they are secondary to the immune-suppression mechanism.

PD-L1: PD-L1 → bTMB axis (Core Path 1) provides a mechanistic bridge linking tumor immunogenicity to adaptive immune suppression via an inflammatory mediator; (b) MSI/dMMR (Core Path 3) and urine-DNA methylation (Core Path 2) inform which tumors are intrinsically more immunogenic or amenable to noninvasive monitoring, but they are secondary to the immune-suppression mechanism.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	bTMB	Supporting modifier from knowledge graph

Secondary	MSI/dMMR	Supporting modifier from knowledge graph
Secondary	urine-DNA methylation biomarkers	Supporting modifier from knowledge graph
Secondary	PD-1	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: provides a mechanistic scaffold to explain this discordance. Secondary modifiers (bTMB, MSI/dMMR, urine-DNA methylation) add context and enable noninvasive monitoring.

Prediction 2: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

Prediction 3: At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- bTMB: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- MSI/dMMR: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.